

Census Income Classification and Customer Segmentation for Marketing

Executive Summary

This project delivers two complementary capabilities for marketing decision-making:

- a **predictive model** to identify individuals likely to earn >50K
- a **segmentation framework** to guide campaign design and messaging

The final CatBoost model achieves strong performance in an imbalanced setting and enables **precision-controlled targeting**. At the selected threshold (0.695), the model captures 32% of high-income individuals while maintaining 88% precision, targeting 2.3% of the population.

In parallel, a **5-persona segmentation framework** organizes the population into distinct life-stage and economic groups, enabling more tailored and effective marketing strategies.

Key insight: effective targeting requires both **who to target (model)** and **how to target (segmentation)**.

1. Exploratory Data Analysis

The dataset contains approximately **199K observations** with demographic, employment, and income-related features, along with a survey weight and a binary income label (>50K).

Two key structural properties of the data directly influence modeling and business decisions:

Class Imbalance (Business Critical)

High-income individuals represent only **~6.2% of the population**, making this a **rare-event classification problem**. As a result, model evaluation must focus on **precision and recall**, rather than overall accuracy.

Heterogeneous Population (Motivates Segmentation)

The population varies significantly across:

- age and life stage
- workforce participation
- education and employment type
- capital income

This heterogeneity indicates that **a single targeting strategy is insufficient**, motivating the need for both:

- a **classification model** (who to target)
- a **segmentation framework** (how to target)

To ensure the analysis remains business-defensible, the survey weight was used to reflect population importance during model training and summary statistics, but it was not treated as a predictive feature. This avoids incorrectly using sampling structure as a customer attribute.

Key Predictive Signals

A small number of features show strong relationships with income:

- **Work intensity (weeks worked)**
- **Capital income (gains, dividends)**
- **Age / career stage**

These findings justify the use of **non-linear models** and provide an economic basis for both model design and segmentation.

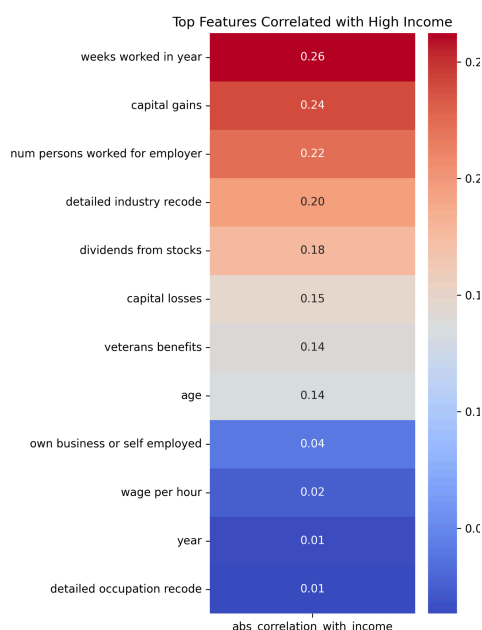


Figure 1. Correlation of top numeric features with high-income status. The chart shows that work intensity and capital-related variables have the strongest association with high income.

Modeling Considerations

- Data was split into train/test with cross-validation for robustness
- Survey weights were used for training and summary statistics, but not as predictive features

2. Classification Results and Recommendation

The dataset presents a **rare-event classification problem** (~6.2% high-income), making precision-recall trade-offs critical.

Three model families were evaluated:

- Logistic Regression (interpretable baseline)
- Random Forest (tree-based benchmark)
- CatBoost (final boosting model)

Model	ROC-AUC	Accuracy	Precision	Recall	F1
Logistic Regression	0.946	0.950	0.792	0.301	0.436
Random Forest	0.947	0.950	0.743	0.336	0.463
CatBoost	0.956	0.954	0.891	0.321	0.472

CatBoost delivers the strongest overall performance, particularly in precision and ROC-AUC, making it the most suitable model for high-confidence targeting in an imbalanced setting.

At the selected threshold (~0.70), the model is tuned for precision-focused targeting, producing a smaller but higher-quality audience, well suited for cost-sensitive or premium campaigns.

2.1 Model Interpretation

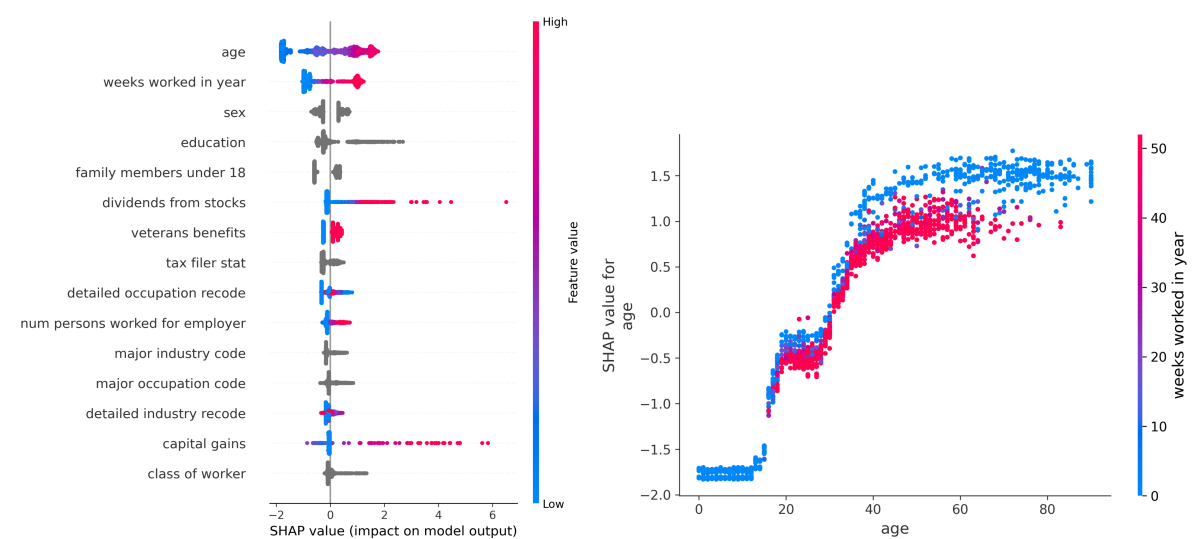


Figure 3. SHAP summary plot of the CatBoost model. Each point represents an individual prediction. Features are ranked by importance, and color indicates feature value (red = high, blue = low).

Figure 4. SHAP dependence plot for age.

SHAP analysis shows that model predictions are driven by a small number of economically meaningful features.

Key drivers include:

- **Age and work intensity (weeks worked)** → proxies for career maturity
- **Capital gains and dividends** → strong indicators of wealth
- **Dependents / low workforce participation** → negative impact

To further examine how the model uses key predictors, Figure 4 shows the SHAP dependence plot for age. The relationship is strongly monotonic: as age increases, the contribution of age to predicted high-income status rises substantially before flattening at older ages. This reflects a life-cycle effect consistent with labor market experience and career maturity. Importantly, the clear color separation reveals a strong interaction between age and weeks worked. For younger individuals, working more weeks is associated with higher predicted income, which aligns with full-time employment. However, for older individuals, those working fewer weeks can exhibit higher SHAP values. This suggests that reduced work intensity at older ages may signal alternative income sources, such as accumulated wealth or capital income. From a business perspective, this highlights that age alone is insufficient for targeting; effective segmentation should jointly consider both life stage and work intensity to distinguish between wage-driven and wealth-driven income profiles.

These feature patterns provide a clear economic basis for the segmentation framework in the next section, particularly in distinguishing younger working adults, mainstream employed households, and smaller affluent niches.

2.2 Recommendation

CatBoost is recommended as the primary model for identifying individuals with income >50K, given its superior performance across all evaluation metrics. Logistic Regression should be retained as a benchmark model for interpretability and validation purposes, but not for production deployment.

2.3 Key Takeaway

The core business question is not whether high-income individuals can be identified. The model demonstrates that this is clearly achievable. The more important decision is how to balance **precision vs. recall**. This trade-off can be directly controlled through threshold selection, allowing the client to align model output with campaign objectives, whether prioritizing high-confidence targeting or broader audience coverage.

3. Segmentation Results and Recommendation

A clustering-based segmentation was developed to group individuals by life stage and economic profile. A 6-cluster solution was selected and consolidated into **5 client-facing segments** for usability.

Segmentation is based on five key dimensions: age, education, occupation, work intensity (weeks worked), and capital gains. These features capture complementary aspects of customer profiles, including life stage, earning potential, and wealth accumulation, enabling meaningful differentiation for campaign targeting and messaging.

Segment	Population Share (%)	High-Income Rate (%)	Avg Age	Avg. Weeks Worked	Avg Capital Gains	Top Education	Top Class of Worker
1: Dependents / non-earners	25.4	0.0	7.8	0.2	0.0	Children	Not in universe
2: Older low-work / retired households	23.2	1.7	55.5	2.5	149.0	High school graduate	Not in universe
3: Mainstream working households	43.0	12.0	39.0	44.9	415.0	High school graduate	Private
4: Younger working adults	8.2	8.0	33.5	43.4	342.0	High school graduate	Private
5: Affluent capital-gains niche	0.2	88.6	46.6	47.4	99,642.0	Bachelors degree (BA AB BS)	Private

Table 1. Client-facing persona summary derived from the selected segmentation framework

- **Segment 1 & 2** → low priority for income-based targeting
- **Segment 3 (Mainstream working)** → core mass-market segment
- **Segment 4 (Younger working)** → growth-focused campaigns
- **Segment 5 (Affluent niche)** → high-value, specialized targeting

Segmentation adds critical context: even high-probability individuals require **different messaging based on life stage and economic structure**.

Recommendation

The client should use the **5-segment framework** for campaign planning. This approach is more actionable than presenting raw technical clusters and better aligned with how marketing teams define strategy, messaging, and channel use.

Key Takeaway

The value of segmentation is not simply that it creates groups, but that it shows why different groups should be approached differently. Even among individuals who may appear economically attractive, campaign effectiveness will depend on matching message and offer to life stage, work profile, and economic structure.

4. Business Recommendations

4.1 Threshold-Based Targeting Options

The model enables flexible targeting depending on campaign cost and objectives.

Scenario	Threshold	Precision	Recall	% of Population Selected	Selected Count	Estimated Cost*
Current Recommendation	0.695	0.881	0.323	2.28%	681	1,362
Balanced	0.500	0.761	0.473	3.86%	1,154	2,308
High Recall	0.300	0.617	0.642	6.46%	1,932	3,864

Table 2. Threshold scenarios for campaign targeting

*Estimated cost assumes a notional contact cost of \$2 per selected individual.

- **0.695** is most appropriate for high-cost campaigns, such as premium outreach or channels where each contact is expensive. It focuses on a small, high-confidence audience.
- **0.500** provides a more balanced strategy, increasing coverage while maintaining reasonably strong precision.
- **0.300** is appropriate only for low-cost campaigns, such as email or digital advertising, where broader reach is more important than targeting purity.

4.2 Recommendation

The current recommended threshold remains **0.695**, because it aligns best with a precision-focused targeting strategy. At this level, the model selects only **2.28%** of the population while ensuring that approximately **88%** of selected individuals are truly high-income.

4.3 Key Takeaway

The model should not be used as a fixed filter, but as a **controllable targeting lever** aligned with budget and campaign goals.

5. Conclusion

Effective targeting requires both **predictive accuracy** and **strategic segmentation**.

The CatBoost model provides a scalable and flexible way to identify high-income prospects, while threshold selection enables direct control over the trade-off between precision, reach, and cost. The segmentation framework complements this by organizing customers into actionable groups based on life stage and economic profile.

Together, these components answer the core business questions: **who to target and how to target them**.

The key takeaway is that targeting effectiveness is not driven by prediction alone, but by combining **data-driven selection with context-aware messaging**, enabling more efficient and strategically aligned marketing decisions.

5.1 Next Steps

- Test score bands instead of relying on one hard threshold
- Validate persona usefulness with campaign response or A/B testing
- Enrich the analysis with client-specific behavioral or transaction data if available
- Calibrate thresholds by campaign economics

Appendix A. Modeling Details

A1. Key EDA Insights (Modeling Motivation)

Feature Relationships with Income

A subset of features shows strong relationships with high-income status, particularly:

- **Work intensity (weeks worked, number of employers)**
- **Capital-related income (gains, dividends)**

Most other variables exhibit relatively weak relationships, indicating that predictive signal is concentrated in a limited set of features. This supports the use of models that capture **non-linear effects and feature interactions**.

Category-Level Differentiation

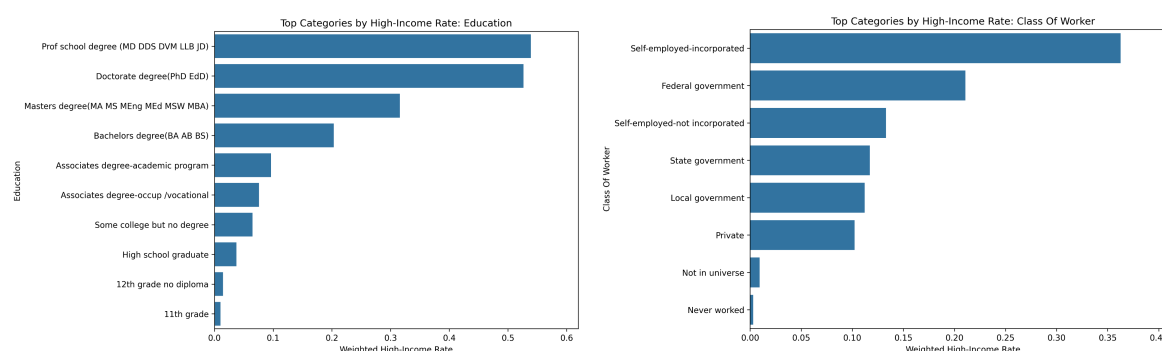


Figure A1. High-income rate by education level and class of worker (weighted)

Education level and employment structure are among the most **interpretable and differentiating variables**:

- Higher education levels → significantly higher income likelihood
- Employment type:
 - **Self-employed incorporated / government roles → higher income**

-
- Private / non-working groups → lower income

Key Implication for Modeling

These patterns directly inform:

- **model selection** (favoring non-linear models)
- **feature importance interpretation (SHAP)**
- **segmentation design**

A2. Data and Preprocessing

Dataset

- ~199K observations with demographic, employment, and income features
- Target: binary income label (>50K vs ≤50K)
- Class imbalance: ~6.2% positive class

Data split

- Train/test split with cross-validation on the training set

Preprocessing Approach (by Model Type)

Different models require different preprocessing strategies:

- **Logistic Regression**
 - One-hot encoding applied to categorical variables
 - Rare-category handling to reduce sparsity
- **Random Forest**
 - Categorical variables encoded numerically (consistent with tree-based requirements)
 - No scaling required due to tree-based structure
- **CatBoost**
 - Native handling of categorical features
 - No explicit encoding required

Survey Weight Handling

- Used as **sample weights** during model training and summary statistics
- Excluded from feature set to avoid treating sampling structure as a predictive signal

A3. Model Evaluation Framework

Models were evaluated using metrics appropriate for imbalanced classification:

- ROC-AUC → overall ranking performance
- Average Precision → performance on rare positive class

-
- Precision / Recall / F1 → targeting trade-off

Threshold selection strategy

- Selected based on **precision-focused objective**
 - Constraint: maintain recall $\approx 0.30+$
 - Aligns with cost-sensitive marketing use case
-

Appendix B. Segmentation Methodology

B1. Clustering Approach

- Method: KMeans clustering
- Evaluated multiple values of k
- Inputs include:
 - age
 - workforce participation
 - capital income
 - education and employment features

B2. Cluster Selection Logic

- **k = 7**
 - Highest internal separation
 - Over-segmentation of mainstream population
 - Created low-value niche cluster
- **k = 6 (selected)**
 - Better balance of separation and interpretability
 - Still required consolidation for business use

B3. Technical to Business Translation

Final output translated into **5 personas** by merging overlapping clusters:

- Reduce redundancy
- Improve interpretability
- Align with marketing use cases

B4. Key Validation Insight

Segmentation results are considered valid because:

- High-income incidence varies significantly across segments
- Key drivers (age, work intensity, capital income) are consistent with model insights
- The affluent niche appears consistently across clustering solutions

Appendix C. Limitations and References

C1. Limitations

- The data is census-derived and may not perfectly match the client's live prospect universe
- Segmentation is unsupervised, so segment value has not yet been validated against campaign response
- The current threshold reflects a precision-oriented objective; other campaign types may justify different cutoffs

C2. References

- Project-provided census data file and column description file
- Scikit-learn documentation for preprocessing, Logistic Regression, model evaluation, and KMeans clustering
- CatBoost documentation for categorical feature handling and classifier training
- Standard references for ROC-AUC and precision-recall evaluation in imbalanced classification